

SNN-IDS: Hardware-Verified SNN-Equivalent Intrusion Detection on a Commodity MCU NPU

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Abstract

We deploy a spiking neural network (SNN)-equivalent intrusion detection system (IDS) on the STM32N6570-DK Neural-ART NPU. By exploiting the mathematical equivalence between single-timestep ($T=1$) SNN inference and INT8 quantized ANN inference, we compile a lightweight MLP (trained on NSL-KDD, 5-class) to a general-purpose microcontroller NPU without any neuromorphic hardware. The resulting INT8 model runs in **0.4561 ms** (2,192 inferences/sec) at 800 MHz, occupying 137.7 KB Flash and 1.25 KB RAM. A comparison experiment with QCFS (Quantization-Clip-Floor-Shift) activation reveals that the `Floor` operator is not implemented in the Neural-ART NPU and falls back to CPU, adding 17.6% latency overhead. To our knowledge, this is the first publicly documented IDS deployment on a general-purpose MCU NPU and the first hardware verification of the $T=1$ SNN-ANN equivalence on commercial NPU silicon. Code and data: <https://github.com/thc1006/SpikeIDS-MCU>

1 Introduction

Deep learning has transformed network intrusion detection, but deploying these models at the network edge remains difficult. Edge IDS must operate under tight power and latency budgets, and the dominant approach today still relies on cloud offloading or high-end GPUs, neither of which fits IoT or telecom CPE (customer premises equipment) scenarios where per-device cost must stay below \$10.

Spiking neural networks offer a different compute paradigm. Because SNNs process information as discrete spikes rather than dense floating-point activations, they can be extremely energy-efficient when run on neuromorphic hardware [9]. The catch is that neuromorphic chips are not commodity parts. Intel Loihi [3] is a research platform. BrainChip Akida is available commercially but carries a price premium. Neither is a standard MCU that an OEM can drop into a low-cost IoT gateway.

A recent line of theoretical work changes this picture. Chen et al. [2] showed that when an SNN runs for exactly one timestep ($T=1$) with zero initial membrane potential, the

threshold-and-fire operation reduces to a ReLU-like clamp, and the forward pass becomes functionally equivalent to an INT8 quantized ANN (under matched quantization grids; the equivalence is approximate in practice due to rounding differences). The practical implication: any NPU that can execute INT8 matrix multiply and ReLU can run an SNN-equivalent model without neuromorphic hardware.

We put this theory to the test. Using the STM32N6570-DK development board (ARM Cortex-M55 at 800 MHz, Neural-ART NPU rated at 600 GOPS INT8), we train an MLP for 5-class intrusion detection on NSL-KDD, quantize it to INT8, and deploy it on the NPU through ST Edge AI Developer Cloud. The model infers in 0.4561 ms with a memory footprint under 140 KB.

We also run a side experiment. The QCFS activation function [1], widely used in ANN-SNN conversion research, should in principle be a better match for SNN semantics than ReLU. We train and export QCFS models and compile them for the same NPU target. The result is a practical finding that, to our knowledge, has not been documented before: the Neural-ART NPU has no hardware support for the `Floor` operator, so each QCFS layer triggers a dequantize $\rightarrow$ floor(float) $\rightarrow$ requantize round-trip on the CPU, costing 17.6% more latency than ReLU despite QCFS achieving 3.3 percentage points higher accuracy.

Our contributions, in summary:

- To our knowledge, we provide the first publicly documented hardware verification of the $T=1$ SNN-ANN equivalence [2] on commercial MCU NPU silicon.
- To our knowledge, we present the first publicly documented IDS deployment on a general-purpose MCU NPU (prior work used FPGAs [7] or neuromorphic chips [13]).
- We document that the `Floor` operator required by QCFS [1] is not supported in hardware by the Neural-ART NPU, identifying a concrete barrier for deploying QCFS-based SNN models on commodity NPUs.

2 Related Work

2.1 SNN-Based Intrusion Detection

Research on applying SNNs to network intrusion detection has grown rapidly since 2024, though nearly all of it remains in simulation. Wang et al. [12] proposed a convolutional SNN for IDS, reporting accuracy above 99% on NSL-KDD in software simulation, but did not address deployment on constrained hardware. Prajwalasimha et al. [7] described an event-driven SNN-IDS architecture targeted at edge IoT devices and tested it on FPGA logic; their results show that spike-based encoding of network traffic features can reduce dynamic power, but they reported no MCU or NPU deployment. Mustafa et al. [6] combined recurrent and spiking layers for anomaly detection in IoT networks, demonstrating that the temporal dynamics of SNNs can capture sequential patterns in traffic data. Most recently, Karthik et al. [4] proposed a transformer-spiking hybrid that fuses protocol-aware attention with spike-coded features. Their model achieves strong accuracy on CIC-IDS benchmarks, but runs only on GPU and makes no attempt at edge deployment.

The only prior work we are aware of that deploys an SNN-based IDS on physical hardware is the CSIAC report by Zahm, Nishibuchi et al. [13], which used the BrainChip Akida AKD1000 neuromorphic processor. They reported 98.4% accuracy at approximately 1 W power. However, Akida is a purpose-built neuromorphic chip with native spike processing; it is not a general-purpose MCU and costs significantly more than the STM32N6 (the Akida development kit lists at \$499 versus \$89 for the STM32N6570-DK). Our work targets the opposite end of the cost spectrum: a commodity MCU NPU that OEMs already use for vision-based products.

2.2 ANN-SNN Conversion

The theory of converting a trained ANN into an equivalent SNN has matured over several stages. Bu et al. [1] introduced QCFS, a quantization-aware activation function that maps ANN activations to discrete spike counts. By training with $QCFS(x) = \lfloor \text{clip}(x/s, 0, L) \rfloor \cdot s$ (where s is a learnable step size and L is the number of levels), the resulting network can be directly converted to a multi-timestep SNN with minimal accuracy loss. PASCAL [8] refined this approach with spike accumulation and adaptive layerwise calibration, achieving near-lossless conversion on ImageNet-scale models.

The most relevant result for our work is the $T=1$ equivalence shown by Chen et al. [2]. They demonstrated that when the number of simulation timesteps is reduced to one, the SNN threshold-and-fire mechanism degenerates to a simple clamp, making the forward pass approximately equivalent to an INT8 quantized ANN. This result eliminates the need for multi-timestep simulation and, critically, means that any hardware capable of INT8 inference can execute SNN-equivalent computations. NeuroFlex [5] has since shown that

co-executing INT8 ANN and SNN layers on the same accelerator can cut the energy-delay product by 57–67%, suggesting that the boundary between “ANN hardware” and “SNN hardware” is blurring.

To our knowledge, this paper provides the first publicly documented empirical validation of the $T=1$ equivalence on commercial NPU silicon. We also test whether QCFS (designed for multi-timestep conversion) offers any advantage when compiled for a $T=1$ NPU target, finding that it does improve accuracy but at a hardware cost that the current NPU architecture cannot absorb.

2.3 Edge AI on MCU NPUs

The STM32N6 is STMicroelectronics’ first MCU with a dedicated neural processing unit. Public deployments on this NPU span computer vision (YOLOv8n, MobileNet, face recognition) and audio processing (YamNet, audio event detection) [10]. The NPU’s hardware operator set covers `Conv`, `Gemm`, `Relu`, `Add`, `Clip`, and `MaxPool` in INT8; operators outside this set fall back to the Cortex-M55 CPU with explicit cache-maintenance overhead for data transfers between the NPU and CPU address spaces [10]. To our knowledge, no prior work has deployed an IDS on any general-purpose MCU NPU. Our work fills this gap.

3 Background

3.1 $T=1$ SNN-ANN Equivalence

A Leaky Integrate-and-Fire (LIF) neuron with membrane potential V , weight matrix $\mathbf{W}$, and threshold θ operates as:

$$V[t] = \beta \cdot V[t-1] + \mathbf{W} \cdot \mathbf{x}[t] \quad (1)$$

$$S[t] = \Theta(V[t] - \theta) \quad (2)$$

where Θ is the Heaviside step function and β is the leak factor. At $T=1$ with zero initial state ($V[0] = 0$), the leak term $\beta \cdot V[0]$ is zero regardless of β , so Eq. 1 simplifies to $V[1] = \mathbf{W} \cdot \mathbf{x}$. The output spike then becomes:

$$S[1] = \Theta(\mathbf{W} \cdot \mathbf{x} - \theta) \quad (3)$$

Chen et al. [2] showed that INT8 post-training quantization (PTQ) produces a discretization that closely matches the threshold-and-fire operation: the quantized ANN output $\hat{y} = \text{clamp}(\lfloor \mathbf{W}\mathbf{x}/s \rfloor, 0, 2^b - 1) \cdot s$ approximates the SNN spike output $S[1] = \Theta(\mathbf{W}\mathbf{x} - \theta)$ when the quantization scale s and threshold θ are matched. In practice, this means any NPU executing INT8 `Gemm` + `Relu` is performing SNN-equivalent computation, with accuracy gaps below 0.5 pp on standard benchmarks [2].

3.2 QCFS Activation

The QCFS activation [1] bridges ANN training and SNN spike encoding:

$$\text{QCFS}(x) = \left\lfloor \text{clip}\left(\frac{x}{s}, 0, L\right) \right\rfloor \cdot s \quad (4)$$

where $s = \theta/L$ is the quantization step, θ is a learnable threshold, and L is the number of quantization levels. Back-propagation through the non-differentiable $\lfloor \cdot \rfloor$ uses a straight-through estimator (STE). After BatchNorm fusion, the ONNX graph contains Gemm, Clip, Floor, and Mul operators.

3.3 STM32N6 Neural-ART NPU

The STM32N6570-DK pairs an ARM Cortex-M55 at 800 MHz with ST’s Neural-ART NPU, rated at 600 GOPS for INT8 workloads. The NPU natively supports Conv, Gemm, Relu, Add, Clip, and MaxPool in INT8. Operators outside this set are dispatched to the CPU, incurring cache-maintenance overhead for NPU $\leftrightarrow$ CPU data transfers [10].

4 Method

Figure 1 shows the end-to-end deployment pipeline. Both paths share the same training and export procedure; they diverge only in the activation function used during training and the resulting ONNX operator graph.

4.1 Model Architecture

We use a four-layer MLP with BatchNorm for 5-class classification (*normal*, *DoS*, *Probe*, *R2L*, *U2R*) on NSL-KDD [11]:

$$\text{Lin}(41 \rightarrow 256) \rightarrow \text{BN} \rightarrow \sigma \rightarrow \text{Lin}(256 \rightarrow 256) \rightarrow \text{BN} \rightarrow \sigma \rightarrow \text{Lin}(256 \rightarrow 128) \rightarrow \text{BN} \rightarrow \sigma \rightarrow \text{Lin}(128 \rightarrow 5)$$

where σ is ReLU (Path B) or QCFS (Path A). The model has 111,365 parameters for ReLU and 111,368 for QCFS (the difference is three learnable thresholds θ , one per hidden layer). At ONNX export, each BatchNorm is algebraically folded into its preceding Linear layer, so the exported graph contains only Gemm+Relu (Path B) or Gemm+Clip+Floor+Mul (Path A).

4.2 Training

Both paths use identical training settings: Adam ($\text{lr} = 10^{-3}$, weight decay 10^{-5}), cosine annealing over 80 epochs, batch size 512. We report results from a single seed (42) throughout; we acknowledge this limits our ability to estimate variance and plan multi-seed runs in future work. We apply inverse-frequency class weighting because the training set is heavily skewed (U2R has only 52 samples out of 125,973; Normal has 67,343). Without class weighting, the model ignores U2R and R2L entirely.

Table 1: Test accuracy (%) on NSL-KDD 5-class classification. ΔAcc is the drop from FP32 to INT8 for the same model.

Model	Prec.	Overall	Macro	ΔAcc
ReLU (Path B)	FP32	76.45	56.32	—
ReLU (Path B)	INT8	76.38	—	−0.07
QCFS $L=4$	FP32	79.75	61.31	—
QCFS $L=4$	INT8	78.05	—	−1.70
QCFS $L=8$	FP32	78.79	59.33	—
QCFS $L=16$	FP32	79.39	60.26	—

For QCFS, we sweep the quantization level $L \in \{4, 8, 16\}$ and report all three. The learnable threshold θ is initialized per-layer from the 95th percentile of activation magnitudes observed on the first training batch.

4.3 INT8 Post-Training Quantization

We quantize FP32 ONNX models to INT8 using ONNX Runtime static quantization with MinMax calibration on 1,000 randomly sampled training instances. Per-tensor quantization (QInt8) is applied to weights and activations.

4.4 NPU Compilation

Models are compiled for the STM32N6570-DK using ST Edge AI Developer Cloud v4.0.0 (compiler: Neural-ART atonn v1.1.3). The compiler partitions the ONNX graph into NPU-executable epochs (hardware or hybrid) and CPU-only epochs (software), generating C code that orchestrates NPU $\leftrightarrow$ CPU handoffs.

5 Experiments

5.1 Dataset

NSL-KDD [11] contains 125,973 training and 22,544 test instances with 41 features. We use the standard 5-class grouping: *normal*, *DoS*, *Probe*, *R2L*, *U2R*. Three categorical features (*protocol_type*, *service*, *flag*) are label-encoded with a combined train+test vocabulary (necessary because the test set contains category values absent from training; this mapping is a deterministic bijection and does not leak label or feature magnitude information). The remaining 38 continuous features are z-score normalized using training-set statistics only.

5.2 Accuracy

Table 1 reports test accuracy for both activation types at FP32 and INT8 precision. QCFS $L=4$ outperforms ReLU by +3.30 pp overall and +4.99 pp macro. We hypothesize that the floor operation during training acts as a form of activation

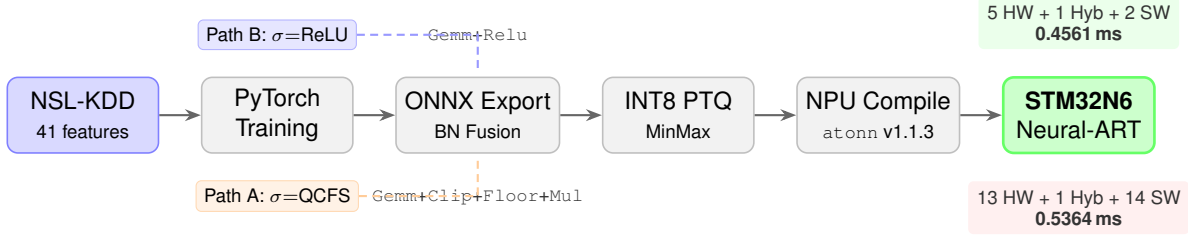


Figure 1: End-to-end deployment pipeline. Both paths train the same MLP on NSL-KDD and export to ONNX with Batch-Norm fusion. Path B (ReLU) produces a Gemm+Relu graph that maps fully to the NPU. Path A (QCFS) adds Floor and Mul operators; Floor falls back to CPU.

Table 2: Per-class test accuracy (%) for FP32 models.

Model	DoS	Probe	R2L	U2R	Normal
ReLU	74.77	62.41	29.70	19.00	95.70
QCFS $L=4$	78.47	76.37	33.26	22.50	95.93

Table 3: Hardware benchmark on STM32N6570-DK (Cortex-M55 @ 800 MHz + Neural-ART NPU). HW = NPU-only epochs, Hyb = NPU+CPU mixed epochs, SW = CPU-only epochs.

Model	Time (ms)	Cycles	HW	Hyb	SW	Flash (KB)
ReLU INT8	0.4561	364,913	5	1	2	137.7
QCFS INT8	0.5364	429,080	13	1	14	138.0
QCFS FP32	1.4156	1,132,485	0	0	20	430.1

regularization, forcing activations onto a discrete grid and reducing co-adaptation between neurons. Consistent with this hypothesis, accuracy generally drops as L increases (79.75 $\rightarrow$ 78.79 $\rightarrow$ 79.39), trending toward the continuous ReLU baseline (76.45). The non-monotonicity between $L=8$ and $L=16$ may reflect interactions with the class-weighting scheme; a more controlled ablation study is needed to confirm the regularization mechanism.

INT8 quantization causes minimal degradation for ReLU (-0.07 pp). ReLU outputs are non-negative, so the effective dynamic range is half that of a signed int8 tensor, leaving more precision per value and reducing quantization error. QCFS suffers a larger drop (-1.70 pp), likely because the floor-based discretization during training imposes a grid that differs from the MinMax calibration grid used by ONNX Runtime’s PTQ, creating a double-quantization effect.

Table 2 and Figure 2 break down accuracy by class. QCFS lifts Probe accuracy by 13.96 pp and R2L by 3.56 pp relative to ReLU. The persistently low U2R accuracy (19–22.5%) is unsurprising given that NSL-KDD contains only 200 U2R test samples and 52 training samples; no amount of class weighting can fully compensate for this level of data scarcity.

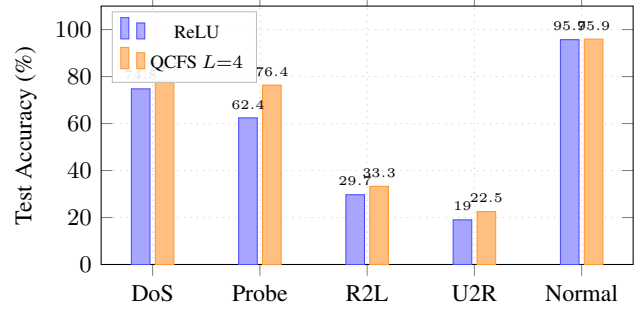


Figure 2: Per-class FP32 test accuracy. QCFS $L=4$ improves all attack classes, with the largest gain on Probe (+13.96 pp). U2R and R2L remain difficult for both models due to extreme class imbalance (52 and 995 training samples, respectively).

5.3.5 NPU Hardware Benchmark

Table 3 and Figure 3 present measurements from the STM32N6570-DK, obtained through ST Edge AI Developer Cloud v4.0.0. The numbers tell a clear story. ReLU INT8 maps cleanly onto the NPU: five hardware epochs handle all Gemm+Relu layers, one hybrid epoch handles the output layer, and only two software epochs are needed for input/output marshalling. Total inference time: 0.4561 ms.

QCFS INT8 runs the same Gemm operations on the NPU (13 HW epochs, reflecting the finer operator granularity from Clip, Floor, Mul decomposition), but each Floor node triggers a CPU fallback. The compiler inserts DequantizeLinear $\rightarrow$ Floor(float) $\rightarrow$ QuantizeLinear at every QCFS layer boundary, because the NPU operates only on integer tensors and Floor is defined on floats in the ONNX specification. This round-trip adds 0.08 ms per inference (+17.6%), distributed across 14 software epochs.

QCFS FP32 runs entirely on the CPU (0 hardware epochs, 20 software epochs) because the Neural-ART NPU only accepts INT8 operators. At 1.4156 ms it is $3.1\times$ slower than ReLU INT8.

Floor operator: a documentation gap. ST’s published operator support lists [10] do not explicitly mention Floor. Our experiment confirms empirically that it is unsupported and quantifies the resulting performance penalty. This finding is relevant to anyone attempting to deploy QCFS-based

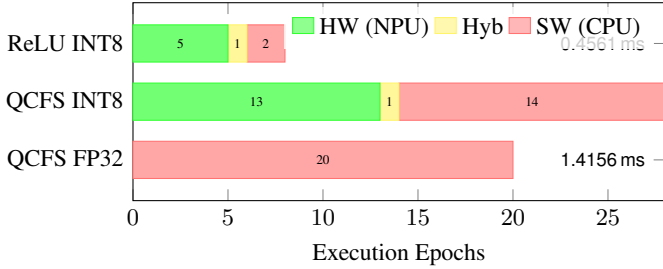


Figure 3: NPU execution epoch breakdown. ReLU INT8 maps almost entirely to NPU hardware (5 HW + 1 Hyb epochs). QCFS INT8 uses 13 HW epochs for Gemm but 14 SW epochs for Floor CPU fallback. QCFS FP32 runs entirely on CPU (20 SW epochs) because the NPU requires INT8 operators.

or other floor-dependent SNN activations on the Neural-ART architecture.

6 Discussion

6.1 Validating the $T=1$ Equivalence in Practice

Chen et al. [2] established the $T=1$ SNN-ANN equivalence theoretically and validated it on standard image benchmarks (CIFAR-10, ImageNet). Our result extends this to a non-vision domain (network security) and, more importantly, to physical NPU hardware. The fact that the INT8 quantized ANN runs on the Neural-ART NPU at sub-millisecond latency with only 0.07 pp accuracy loss confirms that the equivalence is not merely a mathematical curiosity but a deployable engineering technique.

That said, the equivalence is approximate, not exact. INT8 rounding artifacts and the specific quantization calibration scheme (MinMax in our case) introduce noise that differs from the idealized SNN threshold. On NSL-KDD, this noise is negligible. Whether it remains so on higher-dimensional inputs (e.g., raw packet payloads) is an open question.

6.2 QCFS: Better Accuracy, Worse Hardware Fit

The QCFS result presents an uncomfortable trade-off. QCFS $L=4$ beats ReLU by 3.30 pp overall accuracy, which is a meaningful gain for a security application. But the 17.6% latency penalty from Floor CPU fallback makes it a poor choice for this particular NPU.

This is not a fundamental limitation of QCFS; it is a limitation of the Neural-ART operator set. If ST adds Floor support in a future NPU revision (or if a compiler optimization replaces Floor with an equivalent integer shift, as NeuroFlex [5] does in custom accelerators), QCFS could become the preferred activation for SNN-equivalent deployment. Until then, ReLU INT8 is the pragmatic choice.

6.3 Comparison with Akida

Direct comparison with the Akida-based IDS [13] requires caution. The CSIAC report does not fully specify the classification task or dataset split, making accuracy numbers non-comparable. Zahm et al. also used a dedicated neuromorphic processor with native spike support and a different model architecture. Our 76.38% on 5-class NSL-KDD therefore cannot be directly compared to their reported 98.4%. What *is* comparable is the deployment target and cost: we show that a \$89 general-purpose MCU (which OEMs already source for vision products) can run SNN-equivalent IDS inference at 2,192 inferences/sec. In a flow-based IDS pipeline where features are aggregated over connection records (as in NSL-KDD), this translates to classifying over two thousand flows per second, well within the traffic rates of typical IoT gateway deployments.

6.4 Limitations

Four limitations bound our results. First, NSL-KDD is a legacy benchmark derived from 1998 DARPA traffic. It does not contain modern attack types (e.g., DDoS amplification, lateral movement, encrypted C2 channels). We chose it for reproducibility and comparability, not because it represents production traffic. Any deployment based on this work should retrain on contemporary data such as CICIDS2017 or CSE-CIC-IDS2018.

Second, we measure inference latency but not power. The STM32N6570-DK does not expose per-subsystem power rails in a way that allows clean NPU-vs-CPU power attribution. Power measurement requires a custom setup with shunt resistors on the VCore rail, which we leave to future work.

Third, we report accuracy and macro accuracy but not F1-score or precision/recall, which are more informative for IDS evaluation under class imbalance. Future work should report the full confusion matrix.

Fourth, the model is a shallow MLP. Deeper architectures (e.g., 1-D CNN or small transformers) might achieve better accuracy on the minority classes, but would require verifying that all operators still map to the NPU.

6.5 Future Directions

Several directions follow from this work.

Modern datasets. Retraining on CICIDS2017 or CSE-CIC-IDS2018 would test whether the hardware constraints we identified (model size, operator support) hold for more complex traffic distributions.

RTOS integration. The 0.4561 ms inference time fits within a typical 1 ms RTOS tick (e.g., FreeRTOS at `configTICK_RATE_HZ=1000`). A natural next step is to integrate the IDS model into an RTOS task on the STM32N6 (e.g., under FreeRTOS or Eclipse ThreadX), feeding it features extracted from a packet processing pipeline. The Cortex-M55 can handle feature extraction while the NPU runs

inference in parallel.

Power measurement. A proper energy analysis, measuring Joules per inference on both the NPU and CPU paths, would quantify the energy advantage of the $T=1$ approach and allow direct comparison with Akida’s published power figures.

Other MCU NPUs. The NXP i.MX RT1180 (eIQ Neutron NPU), Renesas RA8 (Ethos-U55), and Alif Ensemble (Ethos-U55) all have INT8 NPUs. Testing the same model across these platforms would establish whether the $T=1$ deployment strategy generalizes beyond STM32N6.

Multi-timestep exploration. Although $T=1$ is sufficient for equivalence, running $T>1$ timesteps could be useful if the IDS processes temporal sequences of packets. This would require accumulating membrane potential across timesteps, which the current NPU cannot do in hardware but could be managed in software with reasonable overhead if the timestep count is small (e.g., $T=2$ or $T=4$).

7 Conclusion

We trained an MLP for 5-class intrusion detection on NSL-KDD, quantized it to INT8, and deployed it on the STM32N6570-DK Neural-ART NPU. Inference takes 0.4561 ms. The model fits in 137.7 KB of Flash. These numbers confirm that the $T=1$ SNN-ANN equivalence is not just theory; it works on real, shipping hardware.

The QCFS experiment adds nuance. QCFS trains better (+3.30 pp), but the `FLOOR` operator has no NPU support and forces costly CPU round-trips. For now, ReLU INT8 is the right activation for this class of hardware.

We believe this work opens a practical path for deploying SNN-equivalent models on the large installed base of general-purpose MCU NPUs, bringing neuromorphic-style efficiency to devices that cost under \$10.

All code, trained models, and NPU benchmark logs are publicly available at <https://github.com/thc1006/SpikeIDS-MCU> (DOI: [10.5281/zenodo.18906060](https://doi.org/10.5281/zenodo.18906060)).

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